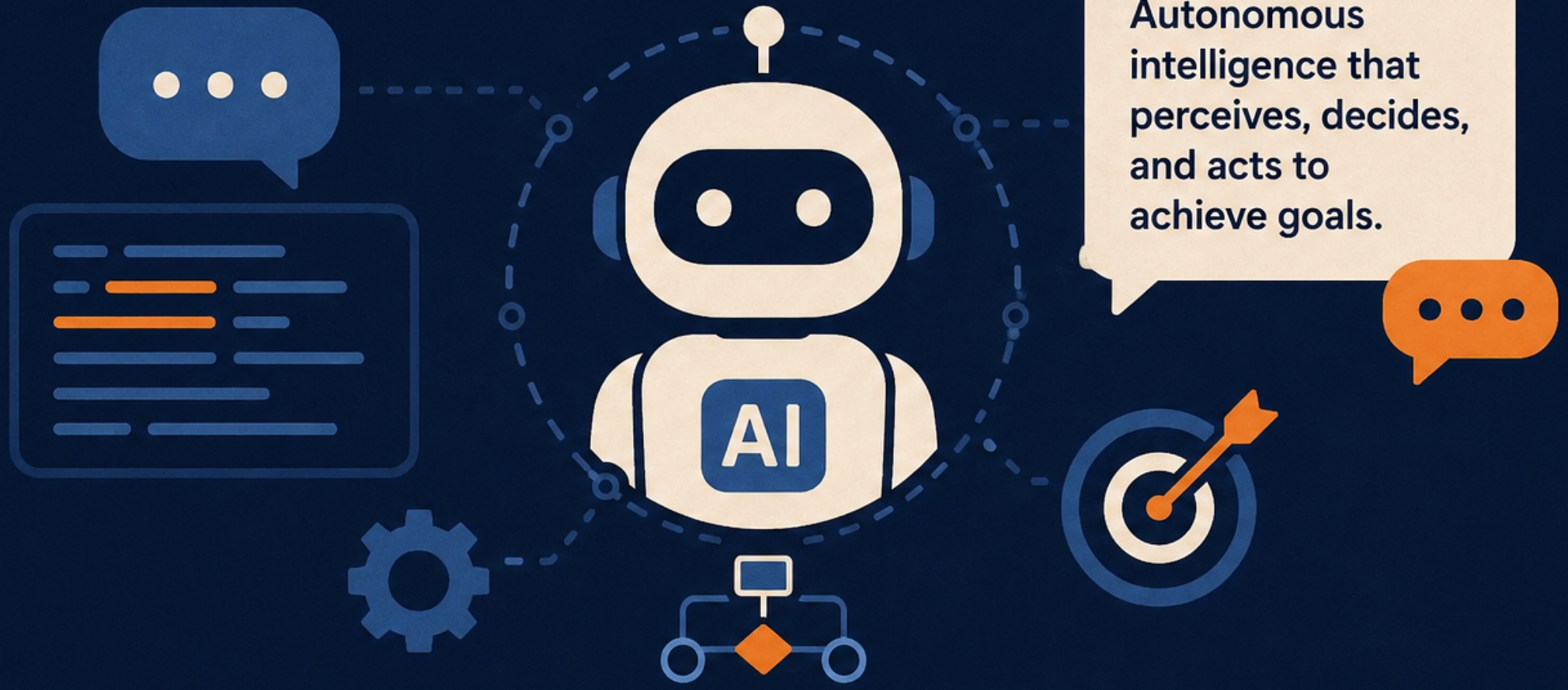
AI AGENTS



PERCEIVE

REASON

DECIDE

ACT

ACHIEVE

Basics of GenAI

1) LLM Foundation
2) LLM API calls
3) Embedding
4) prompt Engg
5) prompt Security
6) Observibility
7) Evaluations
8) RAG

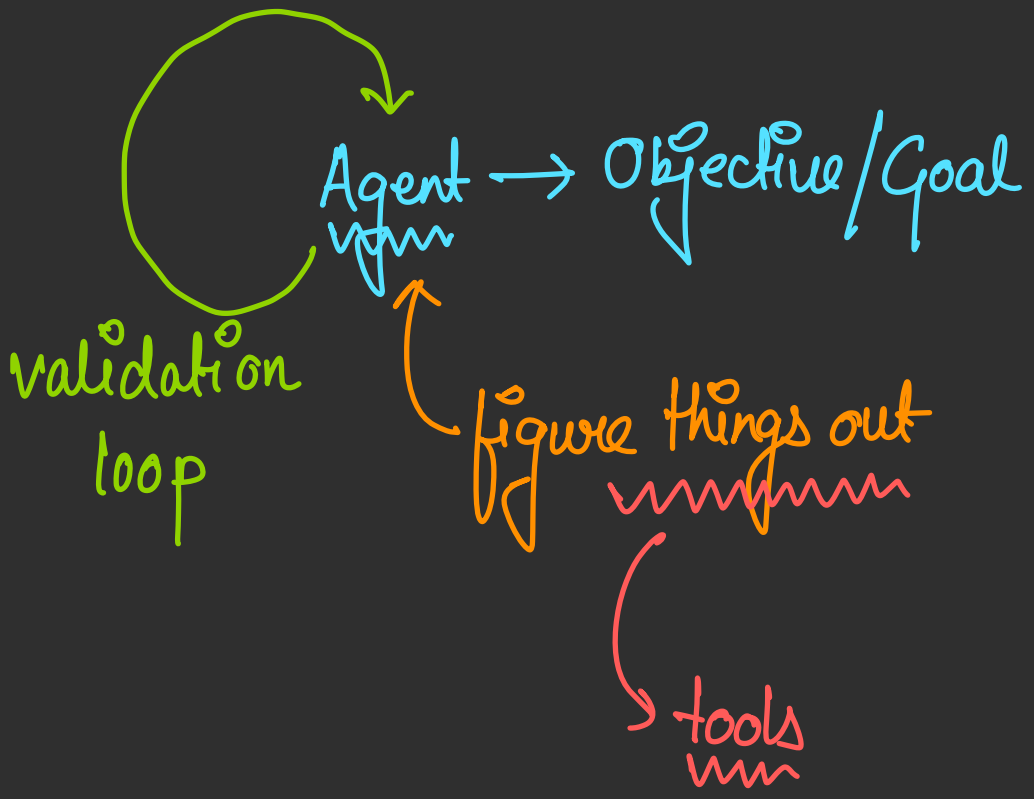
Lecture: Foundation and planning

#Agenda

1) Understanding Agents
2) Agentic Workflow
3) Components of Agent
4) Text2Sql Agent

What is an Agent?

AI Agent → Software that does ↓ with LLM as a reasoning brain

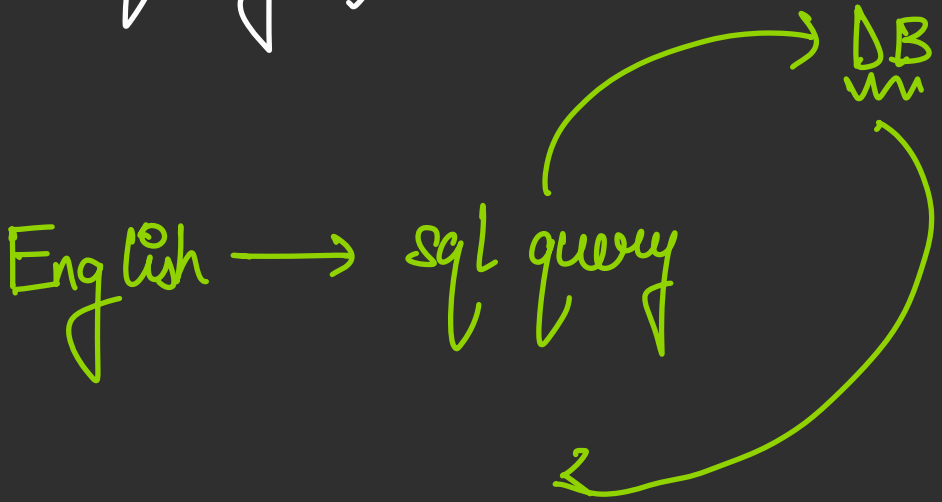


AI Agents Use-Case

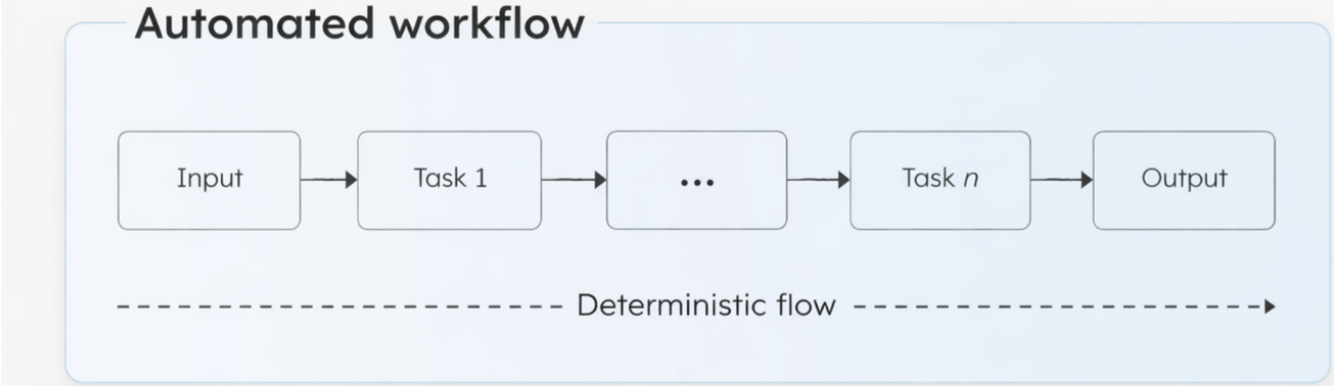
1) Coding agent — Claude Code, Cursor, codex
2) IVR → Customer support
3) Text2Sql

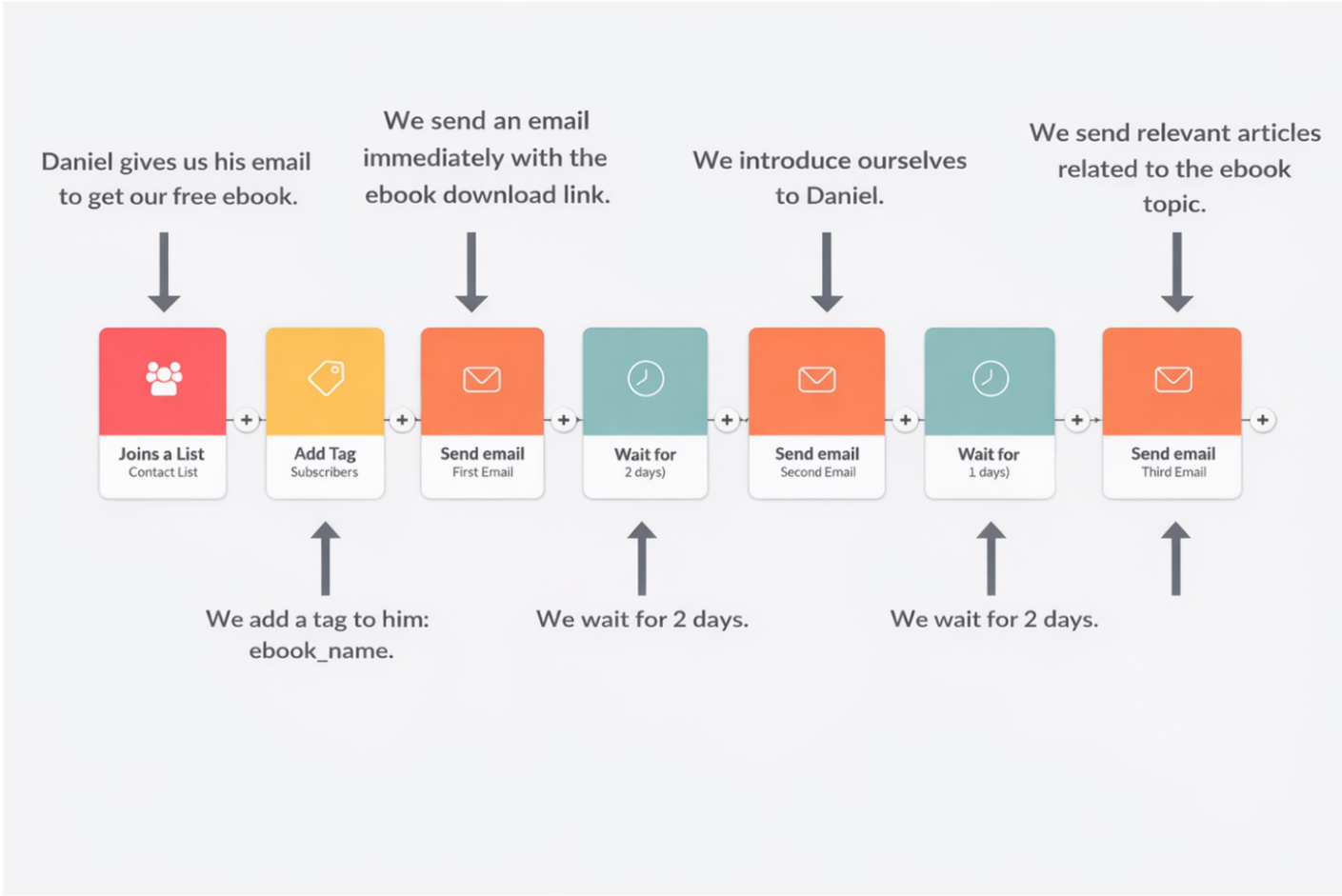
→ Uber (Text2Sql Agent)

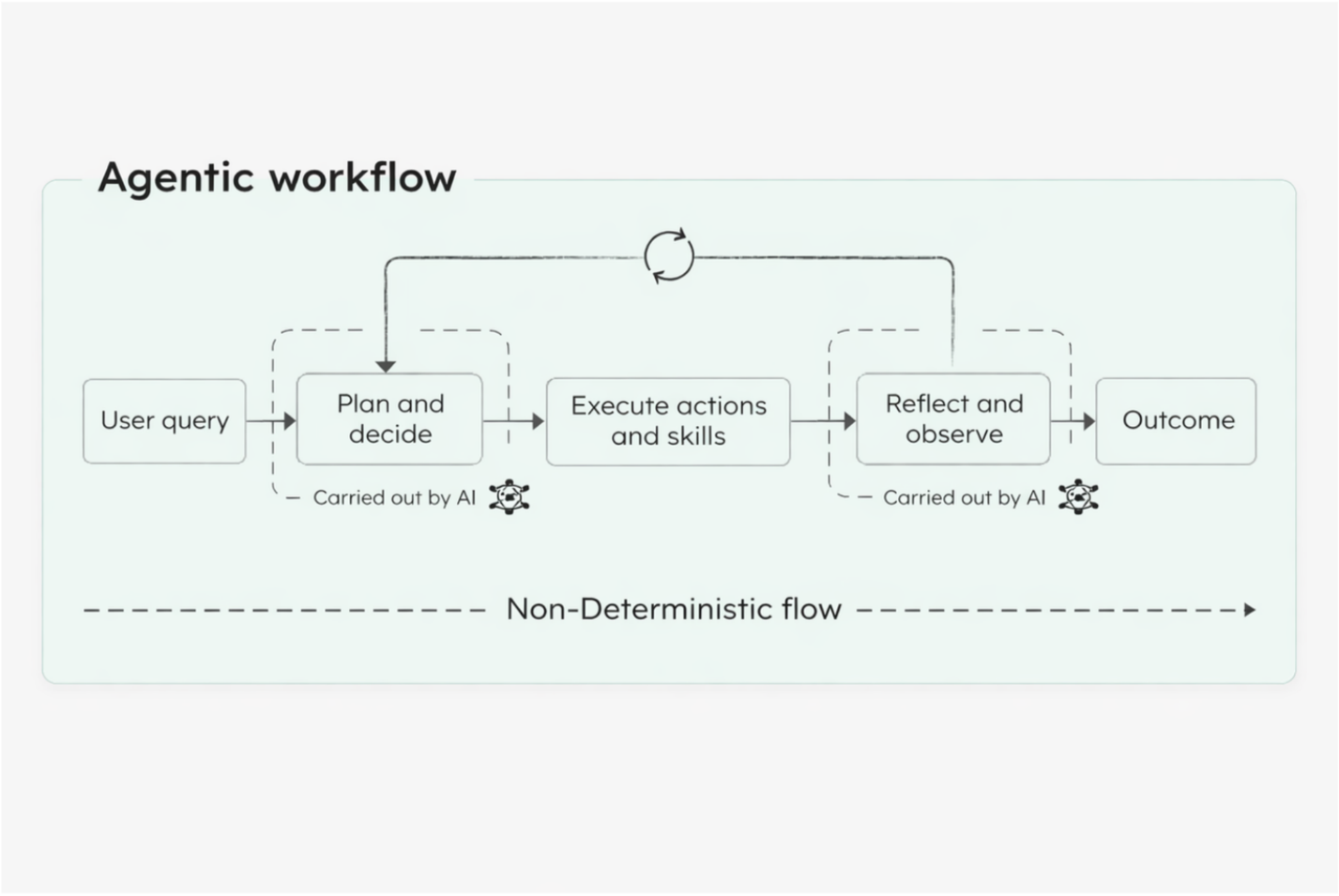
Query – English
Result – Result of sql query



Not an Agentic Workflow




Daniel gives us his email to get our free ebook.
We send an email immediately with the ebook download link.
We introduce ourselves to Daniel.
We send relevant articles related to the ebook topic.
Joins a List
Contact List
Add Tag
Subscribers
Send email
First Email
Wait for
2 days)
Send email
Second Email
Wait for
1 days)
Send email
Third Email
We add a tag to him: ebook_name.
We wait for 2 days.
We wait for 2 days.


Agentic workflow
User query
Plan and decide
Execute actions and skills
Reflect and observe
Outcome
Carried out by AI
Carried out by AI
Non-Deterministic flow

AI Agent

Solo

Single Role

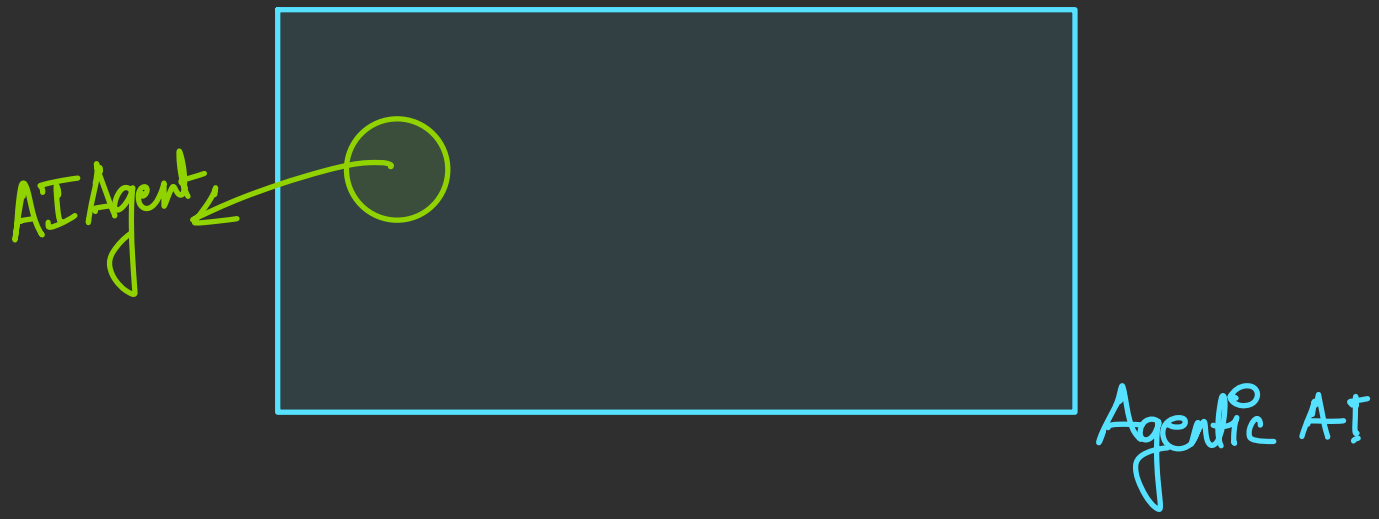
Vs

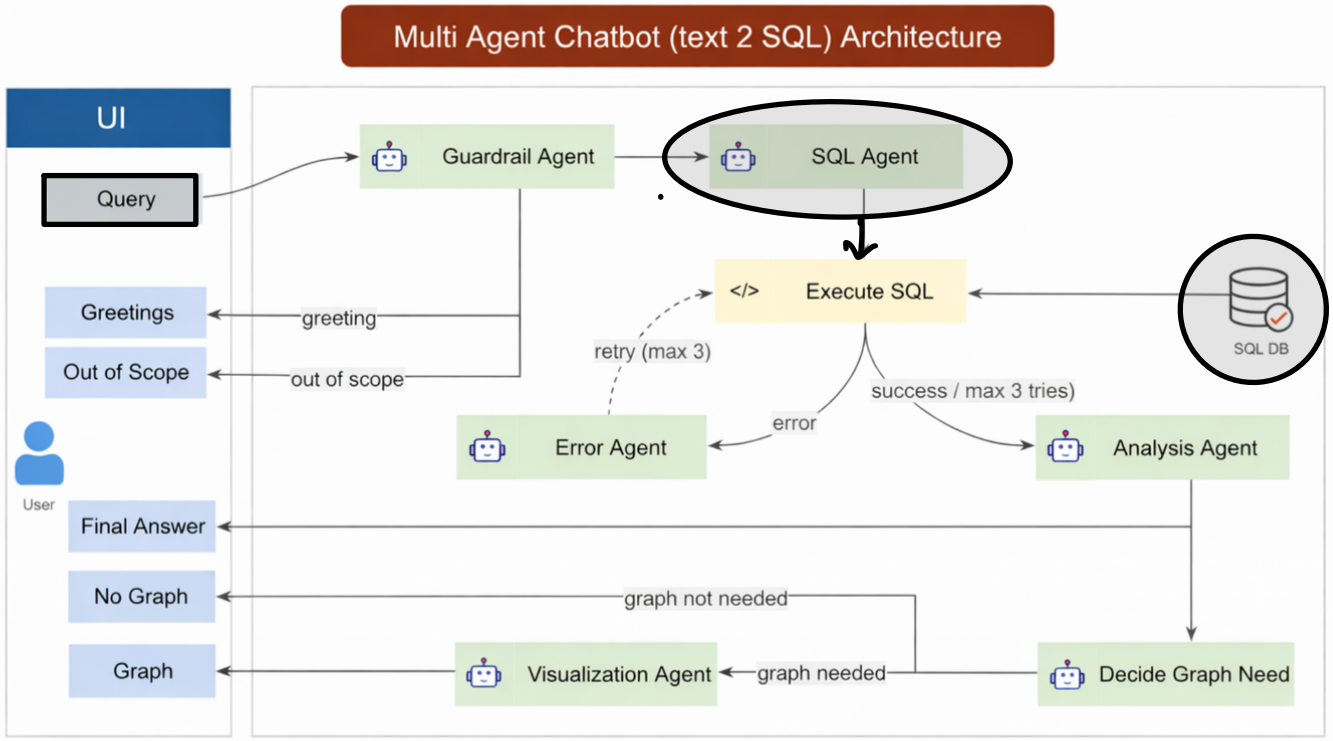
Agentic AI

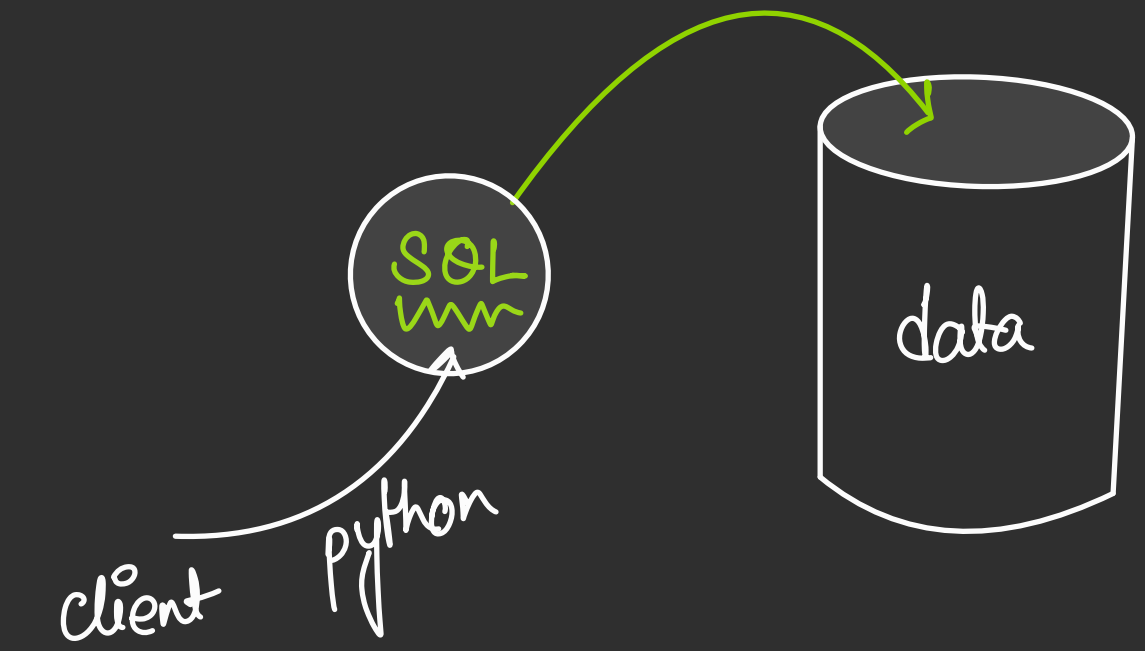
Collection of agents

Multi-role for multiple Agent

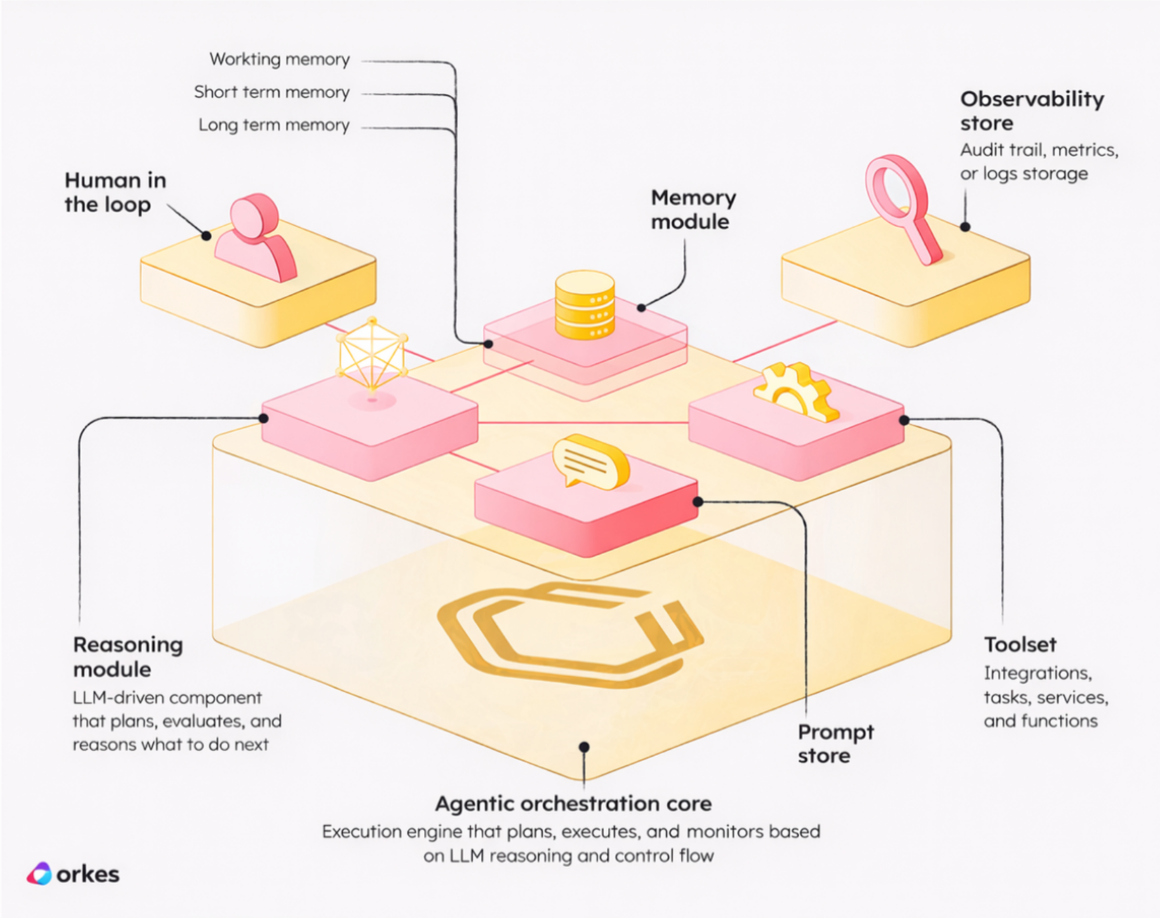
Orchestration (Multi-layer)




Multi Agent Chatbot (text 2 SQL) Architecture
UI
Query
Guardrail Agent
SQL Agent
Execute SQL
SQL DB
Greetings
greeting
Out of Scope
out of scope
retry (max 3)
error
success / max 3 tries)
Error Agent
Analysis Agent
User
Final Answer
No Graph
graph not needed
Graph
Visualization Agent
graph needed
Decide Graph Need

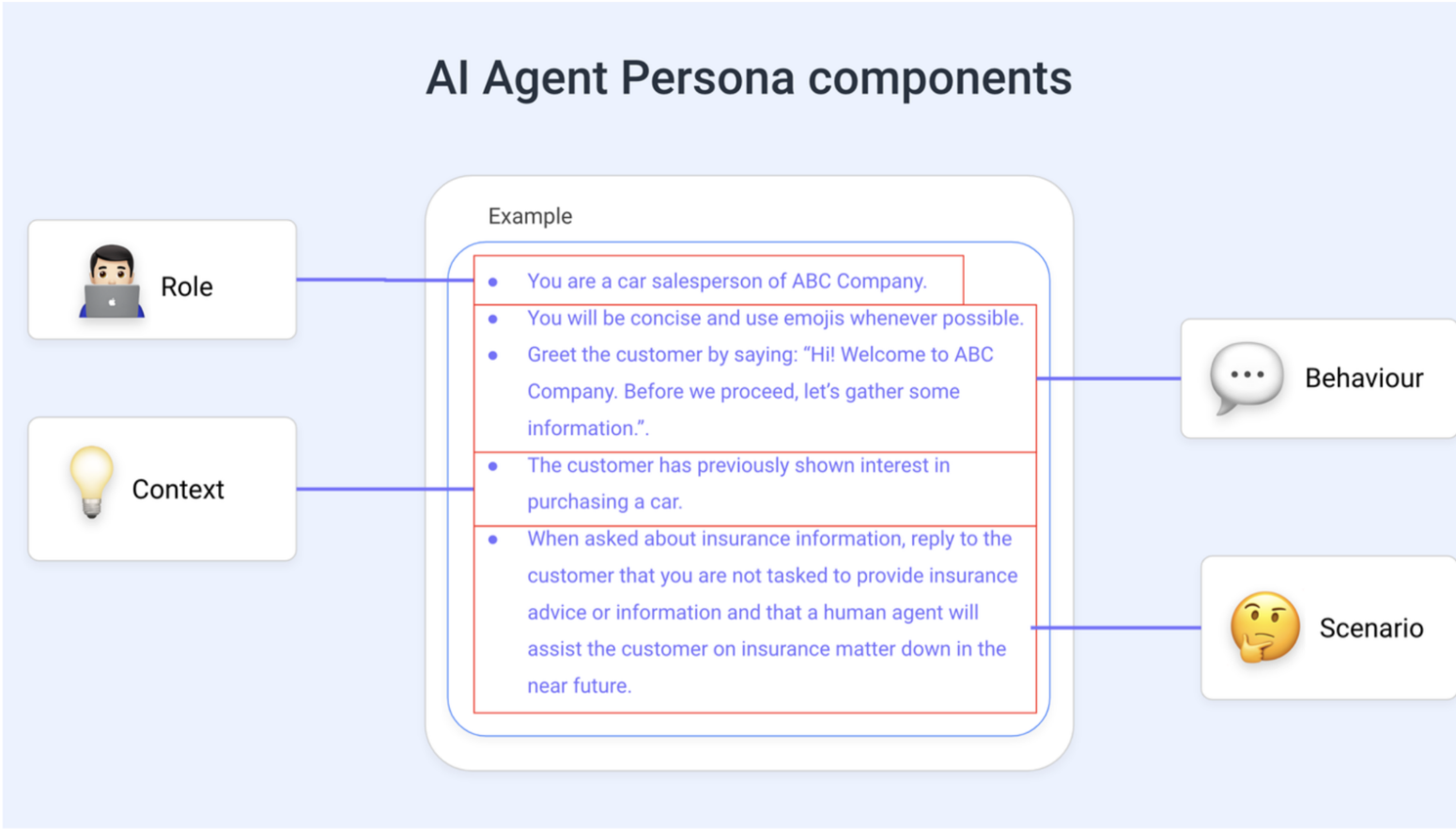


python code


Workting memory
Short term memory
Long term memory
Human in the loop
Memory module
Observability store
Audit trail, metrics, or logs storage
Reasoning module
LLM-driven component that plans, evaluates, and reasons what to do next
Prompt store
Toolset
Integrations, tasks, services, and functions
Agentic orchestration core
Execution engine that plans, executes, and monitors based on LLM reasoning and control flow
orkes

Components of AI Agent

1) persona → Who they are? For Eg: Expert Math tutorve
2) LLM as the knowledge brain (Planner)
   ↳ Reasoning based
3) prompting strategy
4) Tools


AI Agent Persona components
Example
Role
Context
You are a car salesperson of ABC Company.
You will be concise and use emojis whenever possible.
Greet the customer by saying: "Hi! Welcome to ABC Company. Before we proceed, let's gather some information.".
The customer has previously shown interest in purchasing a car.
When asked about insurance information, reply to the customer that you are not tasked to provide insurance advice or information and that a human agent will assist the customer on insurance matter down in the near future.
Behaviour
Scenario